

MITRA PRIVACY TOOL

User Manual

Inspection and Classification of Personal Data Transparency

MITRA Method · TR-Model · LGPD

Guide to using the mitratool.com.br platform and applying the inspection method

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Introduction

The Mitra Privacy Tool is a web application aimed at assessing the level of transparency with which software systems handle personal data. The tool operationalizes the MITRA Method, which organizes the inspection into 46 criteria distributed across five transparency dimensions, drawing on the Brazilian General Data Protection Law (LGPD) and the TR-Model.

This manual is organized as described below. Part I presents the conceptual foundations of the method, so that the evaluator understands what is measured and why. Part II describes the use of each platform feature. Part III, in turn, gathers best practices, a glossary, and references. A reader who intends only to operate the system may begin with Part II; reading Part I, however, improves the inspection and reduces biased interpretations.

Target audience

The target audience comprises developers, data protection officers (DPO), researchers, auditors, and data subjects interested in examining how a piece of software communicates the handling of personal information. Advanced knowledge of design or law is not required, since the method was conceived to allow evaluators with varied backgrounds to carry out systematic analyses.

What the tool addresses

Transparency inspections conducted manually, generally by means of distributed spreadsheets, suffer from the so-called Evaluator Effect, the high variance in results arising from the difference in experience among inspectors, and from the difficulty of consolidating divergent views without resorting to hierarchical power dynamics. To address this, the Mitra Privacy Tool provides contextual cognitive support during completion, mediates the asynchronous resolution of divergences, and generates public accountability artifacts, among them the verifiable badges.

Available languages

The entire interface, the system messages, and the transactional e-mails operate in three languages: Portuguese (pt_BR), English (en), and Spanish (es). The default language is detected automatically from the browser's Accept-Language header and can be changed in the top menu, remaining saved in the user profile.

Foundations of the MITRA Method

1. The MITRA Method

MITRA (Method for the Inspection of Personal Data Transparency) results from the integration between the TR-Model, proposed by Coleti et al. (2020), which defines a minimum set of metadata to promote Personal Data Transparency, and the six Usable Privacy heuristics, proposed by Salgado et al. (2024), which guide privacy-centered and user-experience-centered design. Two main artifacts derive from this integration: a structured checklist, which supports the inspection, and a digital tool, which operationalizes the completion and generates an automated classification.

By adopting the inspection logic based on checklists, similar to classic methods such as ErgoList, MITRA allows specialists or users to carry out systematic analyses even without advanced design knowledge. The checklist can be used independently of the tool, including in printed form.

1.1 Personal Data Transparency

Personal Data Transparency refers to the ability to clearly inform users how their data is collected, used, stored, and shared by applications (Filgueiras et al., 2019). Murmann and Fischer-Hübner (2017) classify transparency into two complementary dimensions: the ex-ante dimension, tied to the right to know in advance how the data will be handled, and the ex-post dimension, associated with the possibility of the user verifying, after use, how their information was actually processed.

Offering transparency in a practical and understandable way requires overcoming recurring barriers, such as translating complex technical content into accessible language, presenting this information clearly in the interfaces, and the absence of consolidated mechanisms that allow assessing, from the user's perspective, the quality of the transparency offered. MITRA acts on this last gap.

1.2 The TR-Model and the Five Dimensions

The TR-Model is a metadata profile that guides the exposure of information about the handling of personal data. Its structure is organized into five classes, presented in the tool as the five transparency dimensions, described below:

Dimension (TR-Model class)	What it describes
People / Actors (<i>Actors</i>)	Who the agents involved in the processing are, such as controller, processor, and recipients, along with their identification and contact data.
Purpose of Use (<i>Purpose of Use</i>)	The purpose for which the data is used, the legal basis that justifies it, and the handling period.
Personal Data (<i>Personal Data</i>)	Which data is collected, its origin, composition (granularity), and the subject's consent.
Sharing (<i>Transfer</i>)	Recipients of the transfers, legal basis, and justification for sharing with third parties.
Agency (<i>Agency</i>)	The subject's rights, the means of contact to exercise them, and the resources available within the software itself.

Figure 1 — The five transparency dimensions and their main inspection elements.

Beyond structuring the content that must be made available, the TR-Model also proposes guidelines for the visual presentation of this information, such as the ideal way to express the granularity of a datum (“Location is composed of Latitude + Longitude + Date + Time”) or to present the legal basis of a purpose of use. MITRA employs the TR-Model as its conceptual basis but, instead of verifying only the presence or absence of the information, guides the evaluator in recognizing, classifying, and scoring the way these elements are presented.

1.3 The Six Usable Privacy Heuristics

Usable Privacy integrates design and usability principles into the construction of systems that handle personal data, with the aim of making privacy practices understandable, controllable, and accessible. Salgado et al. (2024) propose six heuristics, incorporated into MITRA as evaluation criteria, described below:

- **Readability of privacy policies:** policies must be written in clear and accessible language.
- **User doubt and caution:** the design must acknowledge that users weigh risks before sharing data.
- **Offer help and avoid jargon:** technical terms must be avoided or explained with contextual support.
- **Discretionary access control:** control over access and sharing must be visible and configurable.
- **Fast interaction and vulnerability to errors:** interfaces must reduce the risk of error and allow quick configurations.
- **Unstable choices and appropriate symbols:** the design must support changes of preference and employ clear symbols.

In MITRA, these heuristics act as criteria complementary to the TR-Model, since they assess, beyond the informational content, also the way that content is presented to the data subject.

2. The Inspection Checklist

The checklist constitutes the main operational artifact of the method. Each item is composed of four elements, defined as follows:

- **Group:** a thematic set of questions related to transparency.
- **Key question:** a general orientation that guides the inspection within the group.
- **Inspection elements:** specific descriptions of content or form to be verified.
- **Response options:** a limited set of alternatives for recording the assessment.

The questions were organized into thematic groups, with a reduced number of items per category, so as to ease comprehension and reduce cognitive overload during the inspection. The checklist is divided into two complementary groups.

2.1 Group 1 — Existence and Quality of Information

Built from the TR-Model, this group verifies whether the relevant information about data handling is present and whether it is sufficiently clear and complete. The key question that guides it is: “Regarding the volume and quality of the information presented, classify:”. For each element, the evaluator selects one of three options, described below:

Option	Meaning
Sufficient	The information is present and understandable.
Insufficient	The information is present but incomplete, generic, or unclear.
Nonexistent	The information is not available in the application.

2.2 Group 2 — Presentation Format

This group assesses how the information is presented to the user, considering readability, clarity, accessibility, and visual adequacy, based both on the TR-Model guidelines and on the Usable Privacy heuristics. Because it involves perception, interpretive variation among evaluators is admitted, which is why the platform offers deliberation mechanisms, addressed in Part II. The response options are as follows:

Option	Meaning
Appropriate	The presentation is clear, adequate, and easy to locate.
Needs improvement	Part of the presentation is adequate, but there are limitations that compromise the experience.
Inappropriate	The format hinders comprehension of or access to the information.

NOTE · Unified scale in the tool

Although the method names the options differently in each group (Sufficient/Insufficient/Nonexistent in Group 1; Appropriate/Needs improvement/Inappropriate in Group 2), the platform converts them all to a single three-level numeric scale (100, 50, and 0 points), plus the “Not applicable” option. In this way, the score calculation remains consistent across the groups.

3. Scoring and Classification

The metric synthesizes the inspection results into a single, easily interpretable indicator. Each response receives a numeric weight, according to the table below:

Response	Weight
Sufficient / Appropriate	100 points
Insufficient / Needs improvement	50 points
Nonexistent / not answered	0 points

The final score results from the weighted average of the TR-Model classes: the score of each element composes the average of its group; the average of the groups composes the class score; and the average of the classes produces the final score of the application. Based on this value, a symbolic classification is assigned in the form of a medal.

3.1 Medal System

Medal	Range	Interpretation
Gold	91 – 100	High transparency, with clear and complete presentation of the information.
Silver	61 – 90	Moderate transparency: accessible information, but with gaps.
Bronze	41 – 60	Minimal presence of information, with low clarity or completeness.
Incipient	0 – 40	Absence or critical insufficiency of information; requires significant improvements.
IMPORTANT · Classification under development The metric and the classification ranges constitute an initial proposal, defined from a literature review and the authors' judgment. Future studies, involving specialists and methods such as factor analysis or Delphi, should refine it. The result must therefore be interpreted as an orientative diagnosis, without constituting an absolute verdict of compliance.		

4. From Method to Collaborative Platform

The rigorous application of 46 criteria imposes a substantial cognitive load on the evaluation team. When the inspection is conducted collaboratively, the simple mathematical aggregation of the responses (the calculation of averages) tends to conceal critical divergences, which often indicate real ambiguities in the interface under evaluation. To address this, the Mitra Privacy Tool was conceived to create a sensemaking environment that surfaces the interpretive conflict and anchors the assessment in documentary evidence. To this end, the platform acts on three fronts, described in the subsections below.

4.1 Cognitive Scaffolding

To mitigate the Evaluator Effect and inspection fatigue, the interface provides contextual cognitive support through progressive disclosure. For each criterion, the evaluator has immediate access to explicit scoring rubrics, micro-task checklists, and examples of good and bad practices. The design hypothesis is that this scaffolding calibrates the novice evaluator's mental model at the moment of interaction, standardizing the interpretation without requiring extensive training.

4.2 Deliberation Mediation

Instead of forcing an algorithmic consensus, the tool automatically identifies severe divergences (cases in which evaluators assign opposite scores to the same criterion) and highlights them in a dedicated panel, in which peers start asynchronous discussions. The design preserves the evaluator's autonomy, who retains control to change or preserve their own verdict after the debate. In this way, the consolidated result reflects a reasoned deliberation rather than a unilateral hierarchical decision.

4.3 Traceability and Accountability

To address the risk of privacy washing, a situation in which organizations publish empty compliance seals, the tool offers a configurable evidence-requirement mechanism, in which maximum scores may require attaching links or excerpts from the inspected interface. At the end of the round, the platform generates a verifiable badge, which can be embedded in the site of the evaluated software and publicly exposes the assessment metadata. The platform does not semantically audit the evidence, so the final validation of the badge's quality remains under the scrutiny of the data subjects.

Using the mitratool.com.br Platform

5. System Overview

Work on the platform follows a flow from the general to the specific. First, a Project is created, representing the software to be evaluated; next, one or more Evaluation Rounds; and, within each round, the individual Inspections, in which the questionnaire is completed. Upon closing inspections and rounds, the system generates consolidated results, which can be published and turned into badges.



Figure 2 — General flow: from project to verifiable badge.

5.1 Roles and Permissions

Role	Permissions
Owner (<i>owner</i>)	Full control: create, edit, and delete the project, invite members, create rounds, configure rules, and publish results.
Evaluator (<i>evaluator</i>)	Create inspections, complete questionnaires, and view results.
Observer (<i>observer</i>)	View the project and its results only.

6. Registration and Authentication

Create an account

1. On the home page, click **Sign Up**.
2. Enter name, e-mail, and password (minimum of 8 characters).
3. Confirm the password and submit the form.
4. Submitting completes the registration and redirects to the authenticated area.

Log in

1. On the home page, click **Log In**.
2. Enter the registered e-mail and password.
3. If needed, use “Remember me” to keep the session, or password recovery by e-mail.

Edit the profile

The Profile item, in the top menu, allows changing name, e-mail, and password, as well as permanently deleting the account.

7. Workspace

Upon logging in, the main panel leads to the project listing. This screen brings together the projects in which the user participates, as owner, evaluator, or observer, the pending invitations received, with the option to accept or decline, and the button to create a new project.

8. Project Management

A project represents the software to be evaluated and aggregates all its members, rounds, and inspections.

Create a project

1. In the Workspace, click **New Project**.
2. Enter the name (required, referring to the evaluated software), the description, the software URL and, optionally, an icon and a highlight color.
3. Click **Save**. The user is automatically added as the **owner**.

Details screen

The project screen displays the general information (name, description, and URL), the list of members and their roles, the pending invitations, the evaluation rounds, and the inspections. The owner can edit the data at any time, and deletion is reversible (soft delete).

8.1 Advanced project settings

In the Settings tab, the owner parameterizes the behavior and the evaluation rules. These choices determine the rigor of the audit and the way divergences will be handled.

Setting	Description
Evidence Requirement	Requires mandatory textual justification for any response marked as Sufficient (100 points). Recommended in rigorous audits, in which every declared compliance must be accompanied by evidence.
Consensus Model	Defines the resolution strategy when there is a score divergence among evaluators. Detailed in section 10.2.
Audit Type	Self-assessment (internal), conducted by the software's own team, or External Audit (independent), conducted by a third party. The choice generates a distinct provenance badge.
Assessment Visibility	Defines whether the assessments of other members remain visible at any time or stay hidden until completion is finished, so as to preserve the independence of the assessments and avoid confirmation bias.

9. Invitations and Members

Invite members

1. In the members section of the project screen, enter the invitee's e-mail.
2. Select the role: **Evaluator** or **Observer**.
3. Click **Send Invitation**. The system sends an e-mail with an acceptance link, valid for 7 days.

Accept the invitation

If the invitee already has an account, following the link automatically adds them to the project. If not, the link directs them to create an account with the invitation e-mail.

Manage members

The owner can change a member's role between Evaluator and Observer, since the Owner role is not transferable by this means. Invitations can be resent, which renews the deadline, or canceled.

10. Evaluation Rounds

IMPORTANT · Main organizational level

The Evaluation Round groups multiple inspections to generate a consolidated result. It is the unit around which results are published and compared over time.

Create a round

1. On the project screen, click **New Evaluation Round**.
2. Enter a descriptive name, for example, "Q1 2026 Assessment".
3. The round is created with **Draft** status.

Life cycle

Figure 3 — Round life cycle: from draft to consolidated snapshot.

While in draft, the round accepts the creation of inspections. When closed, it generates a consolidated snapshot with the results. It is possible to record the version of the evaluated software, for example, v2.1.3, tying the result to a specific release.

10.1 Close a round

1. On the round screen, click **Review and Close**.
2. The review screen presents a preview of the consolidated result, without saving it yet.
3. Optionally, add a textual diagnosis.
4. If desired, mark it for immediate publication in the Public Directory and choose the visibility level.
5. Confirm to close the round and generate the consolidated snapshot.

10.2 Divergence Panel and Consensus Models

When a round has more than one inspection closed by different evaluators, it is common to have score divergences on certain questions. The system measures the dispersion by the population variance of the scores and classifies the severity of the conflict, according to the table below:

Divergence	Variance
Low	0 to 10
Medium	11 to 30
High	above 30

During review and closing, the Divergence Panel supports the handling of these conflicts according to the consensus model configured in the project, defined as follows:

- **Owner Decides:** the owner views a resolution panel, opens each question in conflict, and definitively sets the consolidated response.
- **Majority Vote:** the system automatically computes the most voted score. Ties are resolved conservatively, with the worst score prevailing, in the hierarchy Nonexistent (0) > Insufficient (50) > Sufficient (100). No manual action is required.
- **Evaluator Convergence:** evaluators discuss each conflicting question in a dedicated chat (conflict thread) and adjust their own scores until consensus is reached.

11. Inspections

An inspection represents the individual assessment of a member regarding the software. Each inspection belongs to a round and to a version of the questionnaire.

Create and conduct an inspection

1. On the project or round screen, click **New Inspection**.
2. Select the round to which the inspection belongs. It is created with **Draft** status.
3. Activate the inspection to enable completion of the questionnaire.
4. When finished, close the inspection. The responses are frozen and the result snapshots are generated automatically.

Figure 4 — Inspection life cycle: draft, active, and closed.

Status	Description
Draft	Newly created inspection, awaiting activation.
Active	The questionnaire can be completed.
Closed	Responses frozen; individual and consolidated snapshots generated.
CAUTION · Irreversible action	
Only the person who created the inspection can activate or close it. Once closed, the inspection cannot be reopened.	

12. Completing the Questionnaire

The questionnaire is based on the MITRA Method and contains 46 questions, organized into two sections that span the five transparency dimensions.

Section	Dimensions covered	Questions
Existence and Quality of Information	People/Actors · Purpose · Personal Data · Sharing · Agency	26
Presentation Format	People/Actors · Purpose · Personal Data · Sharing · Agency	20

Response scale

Response	Value	Meaning
Sufficient	100	The information is present and adequate.
Insufficient	50	The information exists but is incomplete or inadequate.
Nonexistent	0	The information is not provided.
Not applicable	—	The question does not apply to the evaluated context.

How to complete

1. Open the inspection, which must have **Active** status.
2. The questionnaire is presented in cards by section, organized by dimension.
3. For each question, consult the rubric and the examples shown and select the response on the scale.
4. Add an observation justifying the choice, required when the project demands evidence for the Sufficient score.
5. Submit to save the responses.

TIP · Make use of the cognitive scaffolds

Each criterion carries scoring rubrics, examples of good and bad practices, and indications of where to look for the information (footers, banners, and policies). Consulting these resources reduces the variance among evaluators and makes the score more defensible.

13. Results

Upon closing an inspection or a round, the system automatically generates the result snapshots. There are three levels of consolidation, presented in the table below.

Level	What it presents	Where to access
Individual	The result of the personal assessment, with a score per section and per dimension and the final medal.	Inspection → View My Results
Team (consolidated)	Consolidated average of all evaluators' responses in the closed inspection.	Inspection → View Team Result
Round	Consolidation of all inspections in the round, generated upon closing it.	Round → View Round Results

The visualization presents the percentage score per section and global, the corresponding medal, charts per dimension, and the project metadata, such as the questionnaire version and the date.

13.1 Mathematical Transparency of the Calculation

To ensure predictability, the calculation follows deterministic formulas, computed from the bottom up, according to the steps below:

- 1. Base conversion:** the textual responses are mapped to integers (Sufficient = 100, Insufficient = 50, Nonexistent = 0, Not applicable = null).
- 2. Dimension score:** $\text{round}((\text{sum of the questions' points} \div (\text{answered questions} \times 100)) \times 100)$.
- 3. Section score:** simple arithmetic mean of the dimension scores.
- 4. Global score:** simple arithmetic mean of the section scores, with integer commercial rounding, which defines the medal.

14. Evolutionary Comparison

The platform allows comparing results to track the evolution of transparency over time. It is possible to compare two inspections or two rounds of the same project, provided both are closed. The side-by-side visualization highlights improvements and regressions, which proves useful for measuring the impact of corrections implemented between one version of the software and another.

15. Public Directory

TIP · Public access to results

The Public Directory allows anyone, even without logging in, to consult the transparency level of the evaluated software. It is accessible at </tools>.

Publish to the directory

1. Close the evaluation round.
2. On the round screen, click **Publish to Public Directory**.
3. Choose the visibility: **Score Only** (scores and medals) or **Full Report** (questions, consolidated responses, and scores).
4. Publishing generates a unique slug, that is, a friendly URL for the report.

Browse and filter

In the directory, it is possible to filter by medal, year, and questionnaire version, as well as sort by score or by date. Each tool opens its report according to the defined visibility.

Provenance and integrity

When consulting a report, the visitor finds provenance indicators (Provenance Badge): the audit type (self-assessment or external audit), the version of the audited software, the number of evaluators and inspections that composed the average, and the consensus model used.

IMPORTANT · Evaluator privacy

The system automatically sanitizes the public data, removing user identifiers, observations, and comments. The individual assessments of the members are not exposed.

16. Verifiable Badges

After publication, it is possible to generate an embeddable badge to display on the site of the evaluated software. The badge loads the updated data via API, with a five-minute cache, and can be generated in three styles.

Style	Content
Default	Full card: project name, medal, score, and date.
Compact	Compact version: medal and score.
Minimal	Only medal and score inline.

Embed the badge

The badge is inserted into any page by means of a script that references the generated public token:

```
<script src="https://mitratool.com.br/badge/{TOKEN}.js"></script>
```

The public endpoints `GET /badge/{token}` and `GET /badge/{token}.js` return, respectively, the badge data in JSON and the embeddable script. The owner can revoke the badge at any time, making it inactive.

17. Data Export

On the project screen, the Export Project option generates a JSON file with the project and member data, all rounds, inspections, and responses, as well as the corresponding questions, categories, and sections. In Profile, the Export All Data option generates a ZIP file with one JSON per project.

18. Administrative Panel

For system administrators, the panel at /admin offers the management of users, projects, inspections, and invitations, as well as the questionnaire versions, questions, sections, and result snapshots. Access is restricted to administrative profiles and does not interfere with the everyday evaluation work.

Best Practices and Reference

19. Evaluation Best Practices

The recommendations below aim to increase the reliability of the inspection and reduce biases:

- **Involvement of multiple evaluators:** it is recommended that different evaluators complete the questionnaire independently in the same round, since the consolidated result offers a more reliable view than an isolated inspection.
- **Initial independence of the assessments:** the assessments must remain hidden until completion is finished, to prevent one evaluator's score from anchoring the others'.
- **Requirement of evidence for maximum scores:** enabling the evidence requirement for Sufficient responses reduces the risk of declared compliance without documentary backing.
- **Choice of the consensus model according to context:** Majority Vote speeds up extensive rounds; Convergence favors the team's learning; the Owner Decides model concentrates the final word in more controlled audits.
- **Recording of the software version:** the record makes the evolutionary comparison meaningful and ties each result to a specific release.
- **Reading the result as a diagnosis:** the medal guides improvements and does not replace a legal assessment of compliance with the LGPD.

20. Glossary

Term	Definition
Project	Unit that represents the evaluated software and aggregates members, rounds, and inspections.
Round	Grouping of inspections that generates a publishable consolidated result.
Inspection	Individual assessment by a member, tied to a round and a questionnaire version.
Dimension	One of the five TR-Model classes assessed (Actors, Purpose, Data, Sharing, and Agency).
Snapshot	Frozen record of the results, generated upon closing inspections and rounds.
Verifiable badge	Embeddable artifact that publicly exposes the assessment metadata.
Provenance	Indication that the result stems from an internal self-assessment or from an external audit.
Privacy washing	Publication of seals or compliance declarations without real backing.
Evaluator Effect	Variance in inspection results arising from the difference in experience among evaluators.
Cognitive scaffolding	Contextual support (rubrics, examples, and micro-tasks) that calibrates the evaluator during completion.

21. References and Official Sources

Official sources

- TR-Model methodology (USP): https://each.usp.br/cond_met_pand/trmodel/
- Official LGPD portal (Brazilian Federal Government): <https://www.gov.br/anpd/>

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